

**ENGINEERING
PORTFOLIO
2025-2026**

THE KNIGHTS OF NI

TEAM 5206

DECODE
PRESENTED BY RTX

INTRODUCTION

We are the Knights of Ni, a robotics team associated with the Stanford Online High School (OHS), with **team members around the world**. We were founded in the 2011-12 season and have excelled ever since, even qualifying for the Houston World Championship in 2015. This year, we believe that we have accomplished our goals of **building technical knowledge, empowering our community, and building a unique and innovative robot**.



OPERATING AS A GLOBAL TEAM



As a team associated with an **online school**, our members are geographically dispersed across multiple time zones but concentrated in Northern California. Even within NorCal, some members must drive as much as 1.5 hours to attend in-person meetings. Consequently, we meet in an onsite-offsite model, which allows members to contribute remotely.

MEMBERS



RACHAEL
Captain



AANYA
Vice Captain



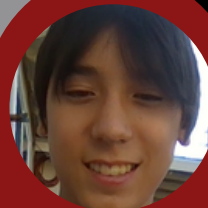
MAXWELL
Software Lead



HAN
Hardware Lead



AYDYN
Outreach Lead



ALEXANDER
Software



WYNN
Software



KATHERINE
Software



ARYAMAN
Software

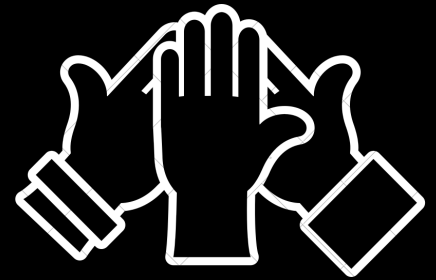


THOMAS
Hardware

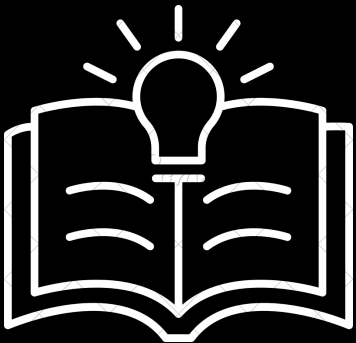
MISSION



EMPOWERING STUDENTS
THROUGH HANDS-ON
ENGINEERING
EXPERIENCES,
PRIORITIZING LEARNING BY
ENCOURAGING RISK-
TAKING, EMBRACING
INCLUSIVITY, AND
FORMING BONDS BETWEEN
MEMBERS



TEAM VALUES



The main objective of our team is **learning**, so we believe it is important that students do as much of the work as possible. Our team hopes to learn and educate through hands-on experience in engineering. This process-oriented approach heavily influences our design decisions; we are more likely to select a unique and interesting mechanism as we prioritize the valuable learning lessons that come with taking these **risks**.

We also pride ourselves as a team open to anyone—it is our policy to not turn anyone away due to lack of experience. Instead, we put a heavier emphasis on **skill-building** through CAD and coding workshops to raise our members to a sufficient level of expertise. Members with varying degrees of experience are encouraged to collaborate with one another, fostering an environment where **inclusivity** and learning work hand in hand.



GOALS



MEASURING SUCCESS

IMPACT OVER OUTCOME

Whether it be inspiring FLL teams or learning a valuable lesson about a new mechanism, we are more appreciative over the journey than an objective success.

STRENGTHENING COMMUNITY

Both within the team and outside, we prioritize robotics as a community, rather than a competition.

INCLUSIVITY

We firmly believe that if anyone has an interest in robotics, it is our mission to cultivate that interest by providing resources.

TEAM ORGANIZATION PLAN

== Team Bonding ==

Virtual movie nights, game nights, and escape rooms

== Onboarding ==

Create a Notion page to introduce new members to the team

== Individual Goals ==

Have members set clear goals for themselves, and help one another achieve them

== Local Community Service ==

Reach out to local libraries and schools for demonstrations, workshops, etc.

== Documentation ==

Increase progress documentation by consistently writing meeting minutes, scheduled deadlines, and task sheets

OUTREACH



As an online team spread across the world, our team sought to engage in virtual and **sustainable** solutions to gaps in our respective **communities** around the globe. We planned outreach events at the intersection of FIRST values and robotics education. We also epitomized Gracious Professionalism on and off the field by **mentoring** other FIRST robotics teams to make robotics more **accessible**. Overall, our outreach efforts **impacted hundreds of students**.

CAD TUTORIALS



We created a **publicly accessible** website with a comprehensive set of Fusion360 tutorials. This website was shared by the Norcal FTC on their website, the NorCal FTC Teams Helping Teams Resource Page.

Our tutorials draw upon our personal experiences working in FTC. They aim to combat the deficits in mentorship, in-person learning, and team collaboration caused by COVID-19 through step-by-step guides and challenge problems that both rookie and veteran teams can take advantage of.

SUPPORT THROUGH DONATION

We donated four boxes of FLL parts to support an FLL team in Austin, Texas. This donation allowed for **an entire FLL team to have individual robots for each member**, allowing students to be able to test ideas individually and then collaborate as a group as well. All together, this allowed the team to be able to work both on teamwork and individual skills.



OUTREACH

STEM COURSE SERIES

Whether it be linear discriminant analysis or the Gram-Schmidt process, our STEM Course Series shares knowledge on a broad variety of STEM topics, like linear algebra. With many members of our team taking **university-level courses, we aim to decrease the barrier for high school students to access such resources.**

IRIS EXAMPLE

Class	μ_{PL}	μ_{PW}	N
Setosa	1.5	0.2	50
Versicolor	4.3	1.3	50
Virginica	5.5	2.0	50

Given the class means, compute the overall mean:

$$\mu = \left[\frac{1}{150} \sum_{i=1}^{150} x_{i,PL} + \frac{1}{150} \sum_{i=1}^{150} x_{i,PW} \right] = [3.76, 1.16]$$

Compute the within-class and between-class scatter matrices:

$$S_W = \begin{bmatrix} 0.22 & 0.08 \\ 0.08 & 0.06 \end{bmatrix}, S_B = \begin{bmatrix} 2.22 & 1.12 \\ 1.12 & 0.62 \end{bmatrix}$$

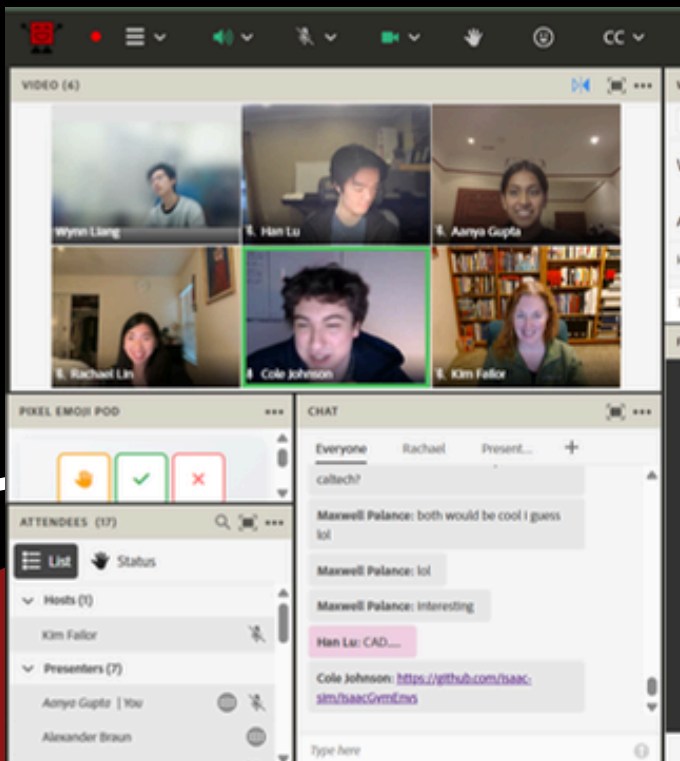
Compute the eigenvectors and corresponding eigenvalues for the scatter matrices:

$$S_W^{-1} S_B w = \lambda w$$
$$S_W^{-1} = \begin{bmatrix} 4.35 & -5.8 \\ -5.8 & 15.9 \end{bmatrix}, S_W^{-1} S_B = \begin{bmatrix} 5.2 & 3.1 \\ 3.1 & 2.6 \end{bmatrix}$$
$$\lambda_1 = 7.1, \lambda_2 = 0.7, w_1 = \begin{bmatrix} 0.8 \\ 0.6 \end{bmatrix}, w_2 = \begin{bmatrix} -0.6 \\ 0.8 \end{bmatrix}$$

iris setosa iris versicolor iris virginica

petal sepal petal sepal petal sepal

SPEAKER SERIES



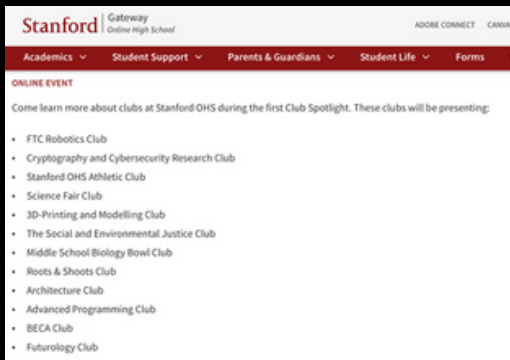
We invited multiple speakers to our school, ranging from individuals in academia to industry. We started by bringing our past vice captain, Thomas Lu, a current physics and computer science student at Berkeley. Thomas **shared the lessons he learned in participating through FTC** and its impact, not only in college, but in his identity.

In addition, we invited Cole Johnson, an alumnus of our school. He is both a researcher and entrepreneur from Georgia Tech and Caltech. From Cole, the team learned a lot about the **applications of robotics** in other fields of study.

OUTREACH



HARDWARE AND SOFTWARE WORKSHOP



Each year, as the anticipation for the season builds, we **host school-wide hardware and software workshops open to all students**, no matter their background in STEM and/or Robotics, to foster interest in robotics and promote FIRST.

MENTORING

HIGH GATE ACADEMY - KIGALI, RWANDA



With international roots, we place a strong emphasis on **global outreach**. Therefore, we took up the opportunity to mentor an FLL Team at High Gate International Academy in Kigali, Rwanda. Each week, we meet with the team to answer any questions they may have, provide guidance on their innovation project, and help them find the same spark we found as young FLL Roboticists. - **49 Hours**

TUTORING

Members tutor 2 **underserved students** K-5 STEM & Humanities. We tutored and designed Python curriculum for **English Language Learners**.



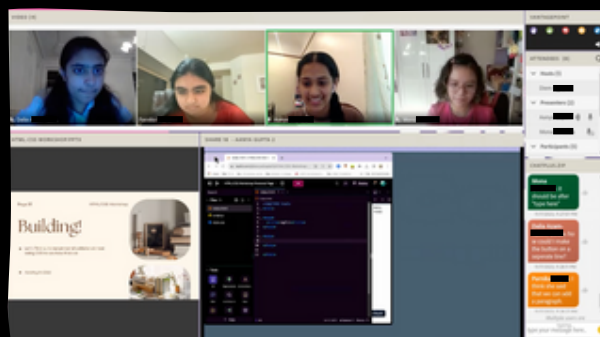
OUTREACH

PEER TUTORS AND TA'S

At Stanford Online High School, **one third of our team members** are peer tutors and/or teaching assistants, including Aanya, Rachael, Aryaman, Alexander, and Maxwell. Peer tutors and teaching assistants are selected at the completion of the previous school year, having exemplified the OHS values of academic achievement and **service to the school community**. They commit at least two hours per week of appointments, as well as hosting weekly office hours and exam review sessions.

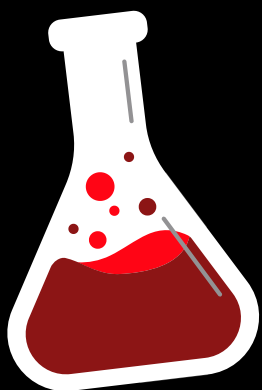
WORKSHOPS

Some of our members are ambassadors for our school's **Girls Can Code Club**, to spread our joy for STEM without communities. This year, we held 2 workshops for students with any coding background. Students learned from the very basics on how to **build websites using HTML/CSS**.



We collaborated with the Madam Math Circle, including students in grades 3-6. We discussed common math terms and theorems applicable in robotics. Further, we guided students through the application of math for this year's FLL mini Mission, Light Show, hoping to make them the next generation of FTC robotics!

CLUBS



Many team members lead or are ambassadors of school clubs, such as the Science Bowl Club, Advanced Programming Club, Logic Olympiad Club, Science Research Club, school magazine, 3D Printing and Modeling Club, and Mathematical Modeling Club. These clubs **promote STEM topics through workshops/speaker series, competitions, and many other events.**

DESIGN PRINCIPLES

Due to our constraints as an online high school, our design team prioritizes five values:

EASE OF ASSEMBLY AND MAINTENANCE

To optimize our time in-person, we aim to minimize maintenance: from choosing a belt-driven viper slide over a string-driven linear slide, to using principles of physics to balance any pressure on servos.

BEING ROBUST

As a team that strives for growth, we ensure our designs are robust—whether that be making efficient uses of space or coupling functions.

RELIABLE

Since drivers have limited time in-person, parts are intuitive to operate. Thus, we can reduce human error. Moreover, by coupling the function of the arm as both intake and outtake, our design's pass-through is incredibly reliable.

Computer-aided design (CAD) is extremely important to us and our entire robot is constructed virtually in the CAD software **Fusion 360 before it is built**. This allows **off-site members to participate in building** and helps us check for interference and other potential issues before the more costly build process begins.

From **7th to 12th graders**, our team features a large range of ages. As a result, we place an emphasis on **mentorship**. To onboard hardware members, we introduce them to the following process:

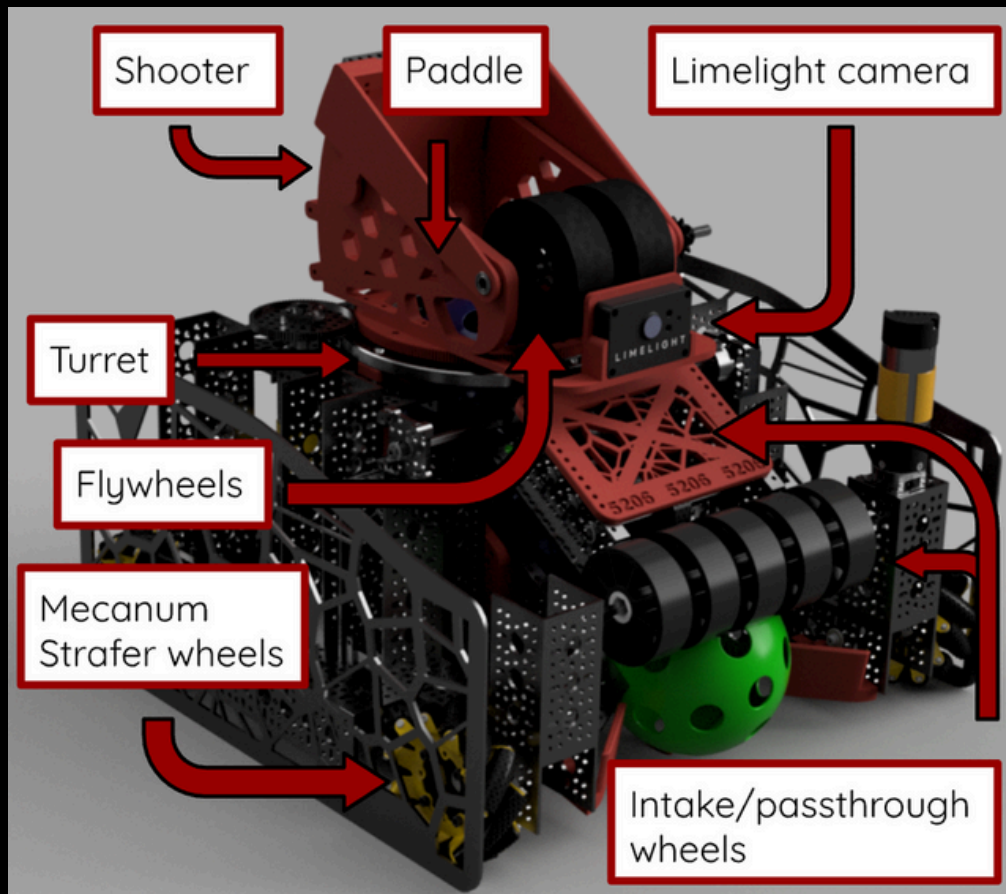
Types of design consideration, such as intake or extension

CAD projects, demos, and tutorials

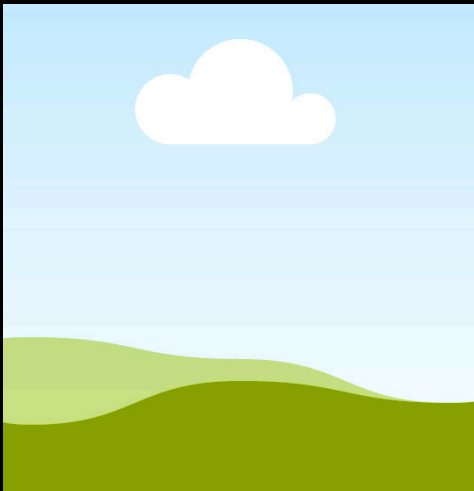
Introduction to 3D printing, using our Prusa MINI+

Training for assembly, maintenance, and iteration

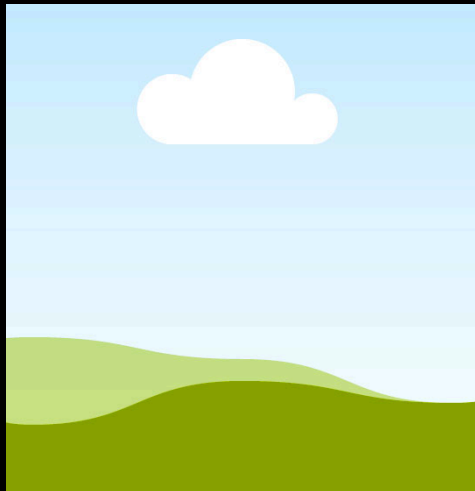
ROBOT OVERVIEW



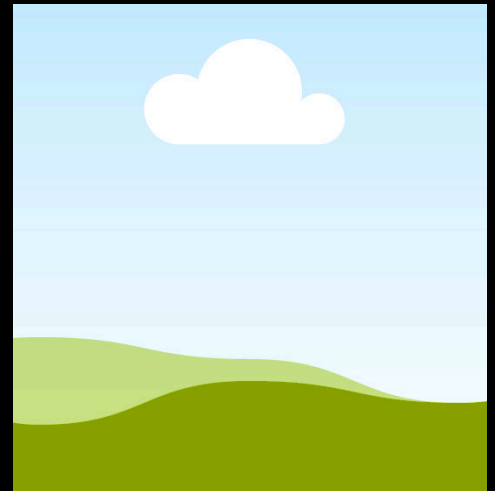
POSITIONS



Shooting Ball 1



Shooting Ball 2



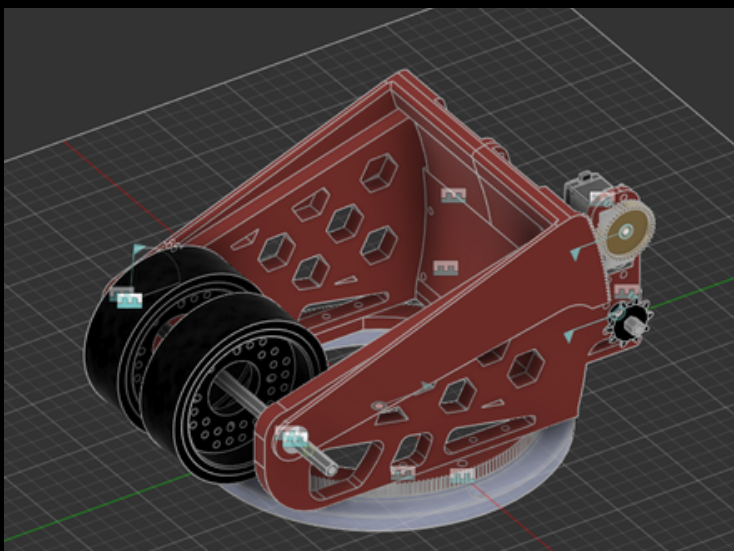
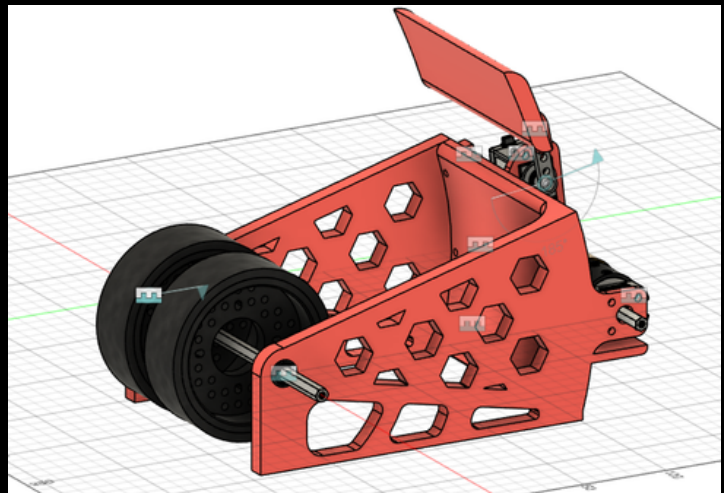
Shooting Ball 3

ITERATIVE PROCESS

Design is an iterative process. Throughout the season, our team **considered a variety of prototypes** and focused on improving on our initial design.

CASE STUDY: SHOOTER

For example, consider the shooter. Initially, our design relied on a shooter which was placed on a rotating turret. However, this design had a few issues such as **not being able to shoot far enough, shooting consistency, and angle consistency**. The flap was initially purposed to vary the angle at which the balls were shot.

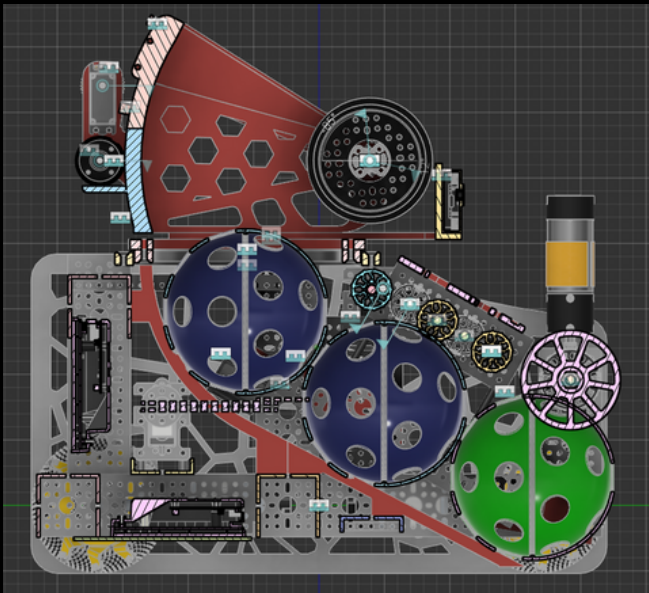


As we iterated on the design, we replaced the initial hood with a new one that had more degrees of freedom. To **improve shooting consistency**, we lined the hood with plastic to **increase traction and friction**, and also added weights on the rotating wheels to **increase momentum**.

INTAKE

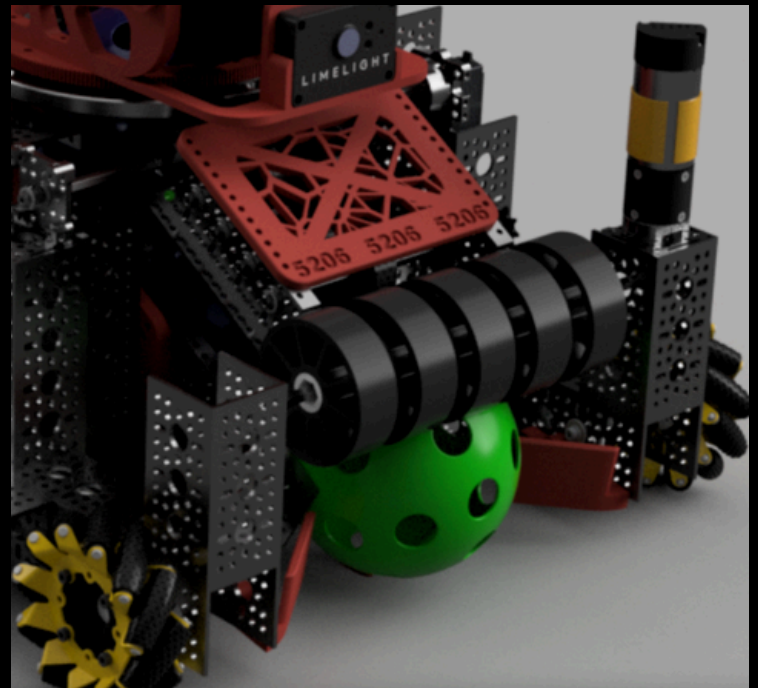
PASSTHROUGH

FINAL DESIGN



Initially, our slope and paddle shooter design **only allowed two artifacts to be shot**, even though we were able to keep three in our robot at a given time. However, this was incredibly inefficient for cycling. Therefore, we **iterated on our design** by adding a second paddle that **allowed us to shoot the last artifact**.

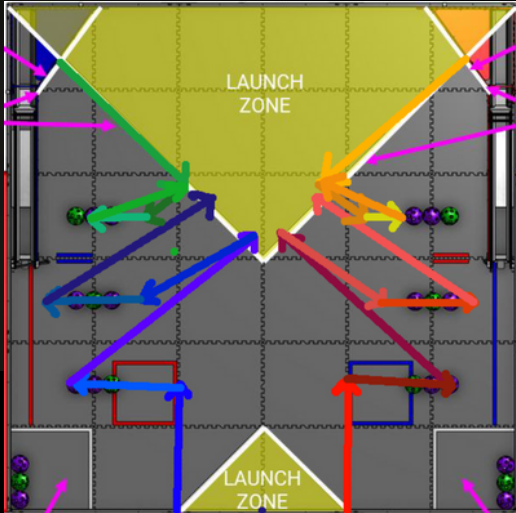
Intaking the artifacts primarily involved driving over them, where the primary set of intake wheels, attached to the robot by an axle, **roll the artifacts into the robot**. The artifact nearest to the turret can then be driven into the turret, where it is fired, by being **pushed by a paddle into the shooter**.



Specifically, let the three artifact storing positions shown in the top image, from left to right, be numbered 1, 2, and 3, respectively. We were only able to shoot artifacts in positions 1 and 2, ending up with the initial artifact in position 3 at position 2. The second paddle brought the artifact at position 2 to position 1, so we could shoot all 3 artifacts at once.

GAME STRATEGY

AUTO



In all of our auto paths, we score samples in the **goals**. We aim to **prioritize iteration** in auto, ensuring the maximum number of elements scored.

In top blue/red we score 3 samples in the goal and then cycle to score another 3 samples. In bottom blue/red we score 3 samples in the goal, pick up three more, and then park.

TELEOP

In TeleOp, our intake collects balls and delivers them to the turret for scoring. The turret uses **Limelight and AprilTag detection to aim automatically**, ensuring smooth and accurate shooting while the driver focuses on positioning the robot. As the distance from the AprilTag increase, the power with which we aim increases.

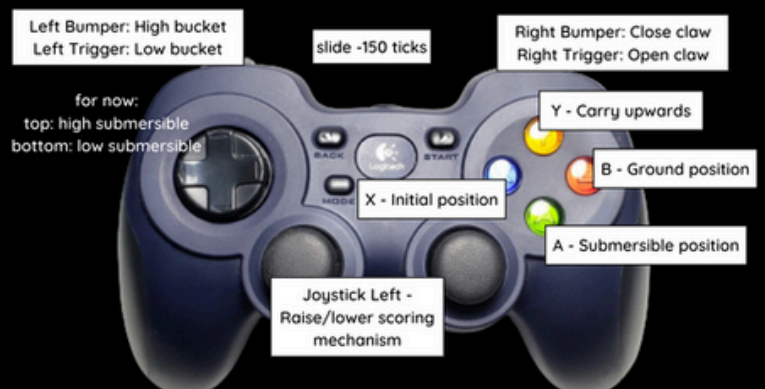


Below is an image of our gamepad mappings. We have coded each gamepad to have the **specific functionality** needed for the drivers. **Drivers work together**, controlling different aspects of the hardware.

Game Control: Primary Driver



Game Control: Secondary Driver

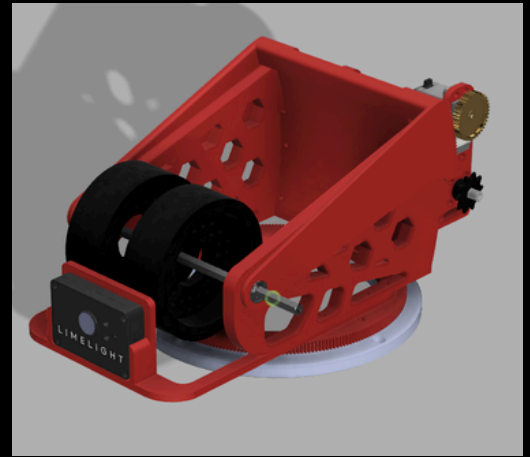


SCORING MECHANISM

LIMELIGHT CAMERA+APRIL TAG

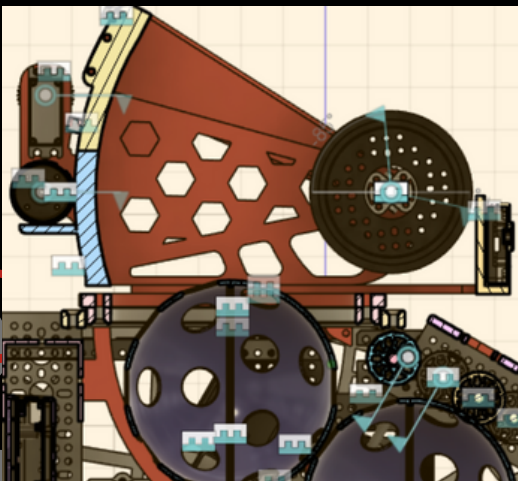
Our hardware design allowed us to utilize the turret through software to track AprilTags, so the driver can more efficiently work on other tasks. The Limelight 3A camera detects nearby tags, and, combined with odometry, **provides accurate pose estimations**.

The turret computes two values, velocity and angle. Angle accuracy is crucial as **even small errors can ruin pose estimate**. We held turret velocity constant during testing and evaluated the turret's reported angle in order to **reduce the amount of independent variables** that make the calculation more murky.



We use a custom-made sigmoid function to smoothly adjust the turret velocity for aiming, rotating the turret until the target is at the center of the camera's field of view, **ensuring the turret works reliably and consistently hits its targets**.

FLYWHEEL



For our robot, our team used a flywheel, which is a spinning wheel used to launch game elements, in this case artifacts, **quickly and consistently**. A fast-spinning mass stores rotational energy (angular momentum), which transfers to the projectile when it contacts the wheel, so we needed to have a fast flywheel **to make our projectiles go high and far**. To find the optimal rotational speed of our flywheel, we tested different motor powers for various distances from the target.

DRIVE

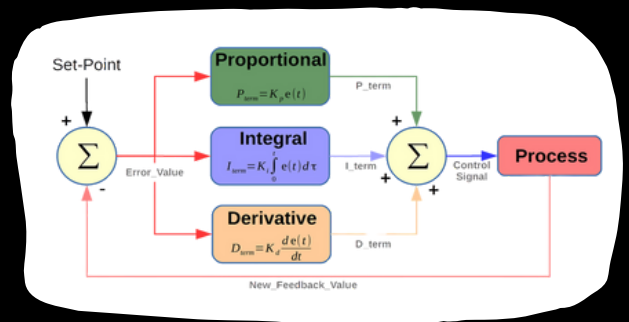
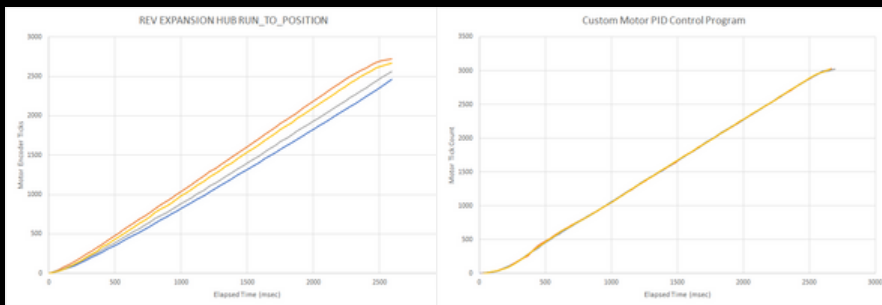
PID

To move our robot consistently and correctly along our selected path, we use PID (Proportional Integral Derivative) control. This aligns our robot's current direction (**kP**), speed (**kD**), and position (**kI**) with our desired states.

- **kP** -> **current error** (where are we currently, where do we want to be?)
- **kI** -> **accumulated error** (what been the outcome of past correction)
- **kD** -> **predicting error** (anticipating future conditions based on current rates of change)

Why do we need PID?

- We found that the robot would not go in a straight line (and still does not), even though we set identical speed and position targets for all four motors.
- Testing revealed inconsistencies in the rate of all the motors, where the REV internal algorithm does not guarantee that all motors rotate synchronously, leading to potential deviations from a straight line.
- PID helps us ensure that all motors move together and addresses inconsistencies, guaranteeing precise movement.
- Before, our current and desired states looked like the graph on the left.



ODOMETRY



This year our robot uses odometry to track its location on the map at all times. This lets us **estimate the position of the robot** by recording the odometry wheels travel along the x and y axes.

The use of odometry helps during the autonomous period as the robot is able to **correct itself** to reach the target location, and from there **more accurately score**.